

Unit 2.5 Limits at infinity

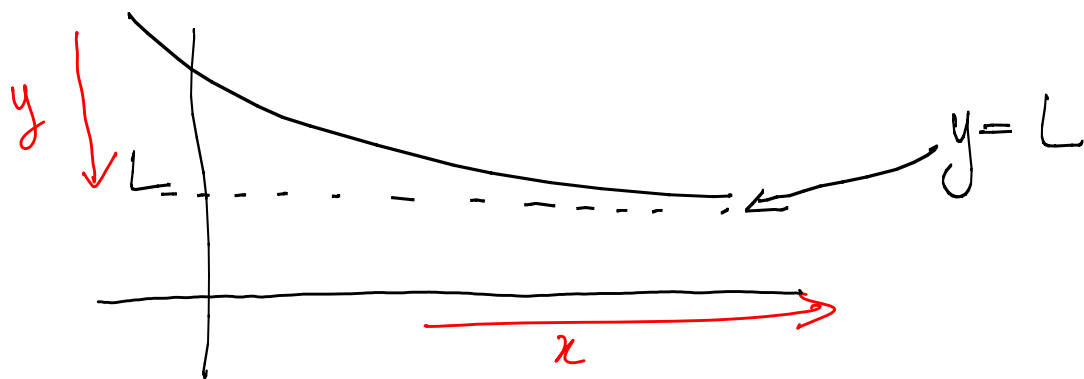
Note Title

2015/03/15

We explore limits of the type $\lim_{x \rightarrow \infty} f(x)$
and $\lim_{x \rightarrow -\infty} f(x)$

Definition: Let f be defined on $(-\infty, \infty)$. Then
 $\lim_{x \rightarrow \infty} f(x) = L$

means that we can bring $f(x)$ arbitrarily close to L
(as close to L as we want) by taking x sufficiently
large.



What is a horizontal asymptote?

$y = L$ is called a horizontal asymptote of f
if either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

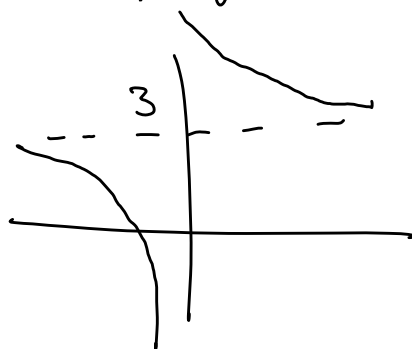
Examples:

1. Find the horizontal asymptotes of $f(x) = \frac{1}{x} + 3$

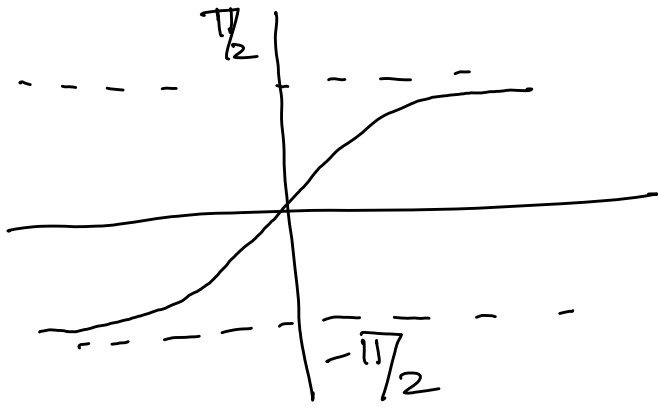
$$\lim_{x \rightarrow \infty} \left(\frac{1}{x} + 3 \right) = 3$$

$$\lim_{x \rightarrow -\infty} \left(\frac{1}{x} + 3 \right) = 3$$

$\Rightarrow y = 3$ is a HA.



2. Find the horizontal asymptotes of $f(x) = \arctan x$.



$$\lim_{x \rightarrow \infty} f(x) = \frac{\pi}{2}$$

$$\lim_{x \rightarrow -\infty} f(x) = -\frac{\pi}{2}$$

The function has two HAs
at $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$.

3. Find: $\lim_{x \rightarrow \infty} \frac{1}{\sqrt[3]{x}} = 0$

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt[3]{x}} = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{x}} \text{ does not exist.}$$

$$\left(\frac{1}{\infty} \right) \downarrow 0.$$

$$\left(\frac{1}{\infty} \right)$$

$$\frac{1}{x^{1/2}}$$

NB.

In general:

If $r > 0$ is rational then $\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$

If $r > 0$ is rational and $\frac{1}{x^r}$ is defined then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0.$$

Eg $\lim_{x \rightarrow \infty} \frac{1}{x^5} = 0$.
rational

$$\lim_{x \rightarrow -\infty} \frac{1}{\sqrt[4]{x}} \text{ Does not exist.}$$

Limits at infinity of rational functions

$$f(x) = \frac{Q(x)}{P(x)} \quad \begin{array}{l} \text{polynomial} \\ \text{polynomial} \end{array} \quad \text{eg } f(x) = \frac{x^3 + 2x}{x^5 + 1}$$

Eg. 1. Find

$$\lim_{x \rightarrow \infty}$$

$$\frac{2x^4 - x^3}{4x^4 + 2}$$

$$= \lim_{x \rightarrow \infty}$$

$$\frac{2 - \frac{1}{x}}{4 + \frac{2}{x^4}}$$

$$= \frac{1}{2}$$

$\Rightarrow y = 1/2$ is a HA.

So if degree (Q(x)) = degree (P(x)) then $f(x)$ has a horizontal asymptote

$$2. \quad \lim_{x \rightarrow \infty} \frac{x^2 - 3x}{x^3 + 1}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} - \frac{3}{x^2}}{1 + \frac{1}{x^3}} = 0.$$

$$\left(\frac{\infty}{\infty} \right)$$

$$\left(\frac{0}{1} \right)$$

$\Rightarrow y = 0$ is a HA

If degree (Q(x)) < degree (P(x)) then $y = 0$ is a HA.

$$\begin{aligned}
 3. \quad & \lim_{x \rightarrow \infty} \frac{x^3 + x + 1}{x^2 + 1} && \left(\frac{\infty}{\infty} \right) \\
 &= \lim_{x \rightarrow \infty} \frac{x + \frac{1}{x} + \frac{1}{x^2}}{1 + \frac{1}{x^2}} && \left(\frac{\infty}{1} \right) \\
 &= \infty
 \end{aligned}$$

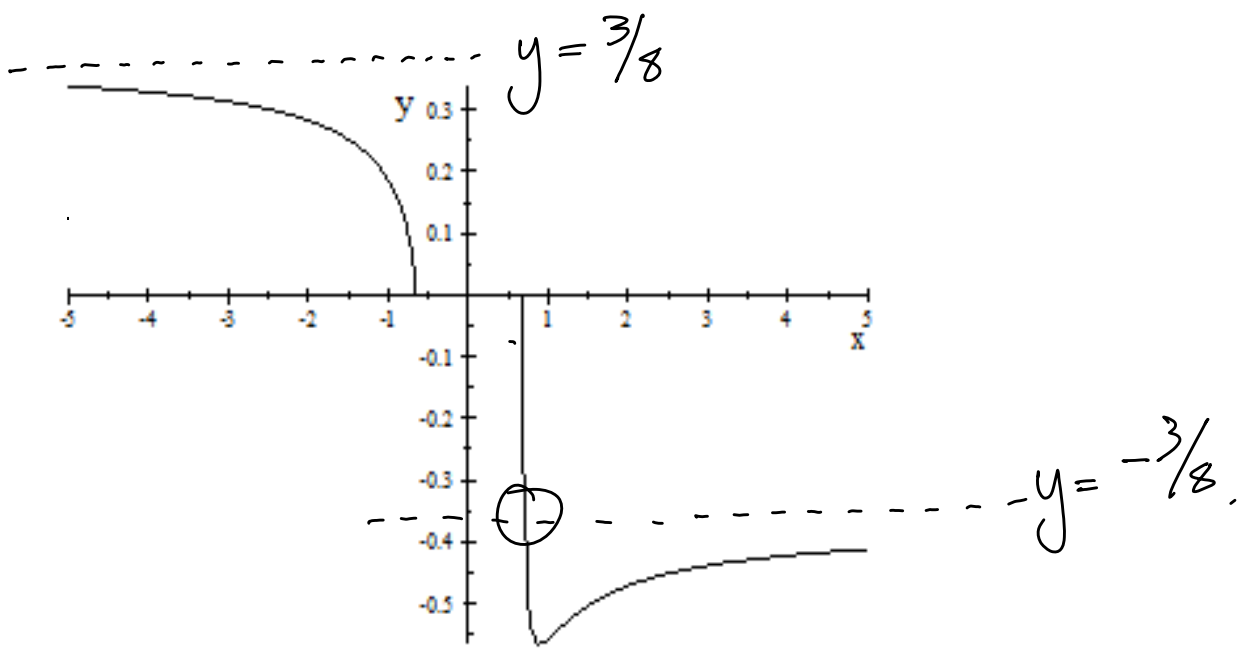
If degree $Q(x) > \text{degree } P(x)$ then there is no H.A.

Important example :

Find the horizontal asymptotes of

$$f(x) = \frac{\sqrt{9x^2 - 4}}{4 - 8x}$$

$$\begin{aligned}
 & 9x^2 - 4 > 0 \\
 \Rightarrow & x > \frac{2}{3} \text{ or } \\
 & x < -\frac{2}{3}
 \end{aligned}$$



Recall that $\sqrt{x^2} = |x|$.

$$(\sqrt{(-5)^2} = 5).$$

$$f(x) = \frac{\sqrt{9x^2 - 4}}{4 - 8x}$$

Recognize

$$\sqrt{9x^2 - 4} = |x| \sqrt{9 - \frac{4}{x^2}}$$

UB.

$$\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 - 4}}{4 - 8x}$$

$$= \lim_{x \rightarrow \infty} \frac{|x| \sqrt{9 - \frac{4}{x^2}}}{4 - 8x} \quad \text{Trick.}$$

$$\sqrt{x^2} = |x|$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{|x|}{x} \sqrt{9 - \frac{4}{x^2}}}{\frac{4}{x} - 8}$$

$$= \frac{3}{-8}$$

$$\Rightarrow y = -3/8 \text{ is a HA.}$$

$\rightarrow -1$

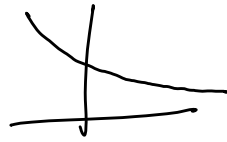
$$\lim_{x \rightarrow -\infty} \frac{\frac{|x|}{x} \sqrt{9 - \frac{4}{x^2}}}{\frac{4}{x} - 8}$$

$$= \frac{\sqrt{9}}{8} = \frac{3}{8}$$

$$\Rightarrow y = 3/8 \text{ is a HA.}$$

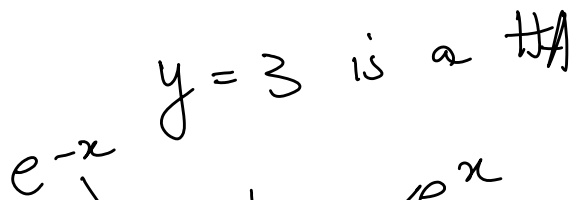
Well-known functions

$$f(x) = e^{-x}$$



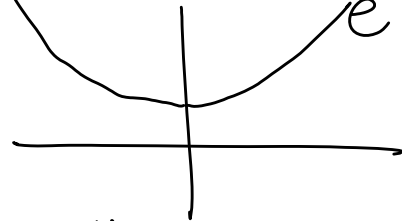
$y=0$ is a H.A.

$$f(x) = e^x + 3$$



$y=3$ is a H.A.

$$f(x) = e^{|x|}$$



No H.A.s.

$$f(x) = \ln x$$

No H.A.s

VA $x=0$.

$$f(x) = \arcsin x$$

